USEFUL EQUATIONS and CONVERSION FACTORS for the QEP and GES EXAMINATIONS

The following equations and conversion factors will be available on the computer used to take the QEP and/or the GES exam. No assurance is given that this list is complete or that the proper use of this information will assure the successful completion of either examination. No attempt has been made to standardize variables. There may be information on this sheet that is not reflected in the GES or QEP exams.

I. Constants and Conversion Factors

°F=9/5(°C)+32

K=°C+273.15

Molar volume at 25°C, 1 atm=24.45L

$1ft^3$=28.32L

1 ft^3=7.481 U.S. gal

1L=1.0567 qt [RAV comment: 1.0567 qt]

1 inch=2.54 cm

12 inches=1 foot

1 lb=453.6 grams

1 gram=15.43 grains

1 atm=14.7 psi=760mm Hg=29.92 in Hg=33.93 ft water=1013.25 mbar=101,325 pascals

1 BTU=1054.8 joules=0.293 watt hr

1cal=4.184 joules

Avogadro's number=$6.024x\ 10^{23}$

gas constant, R=8.314 J/mole K=0.082 L atm/mole K

density of air=1.29 g/L at atm, 0°C

g=981 cm/sec^2=32 ft/sec^2

1 gallon of water = 8.34 pounds

pi = π = 3.142

II. COMMON ENVIRONMENTAL FORMULAS

a. General/Cross Discipline

Area

Rectangle or Square: A = L x W

Circle: A= 0.785 x D^2

Triangle: A= $\frac{B \text{ x } H}{2}$

Area of an open-ended cylinder = $2\pi rh + \pi r^2$

Area of a close-ended cylinder= $2\pi rh + 2\pi r^2$

Volume

Rectangle: Volume = Area x Height

Volume = L x W x H

Cylinder: Volume = Area x Height

Volume = 0.785 x D^2 x H

Cone: Volume = $\frac{\text{Area x Height}}{3}$

Volume = $\frac{0.785 \text{ x } D^2 \text{x H}}{3}$

Velocity

Velocity = $\frac{\text{Distance traveled}}{\text{Time}}$

Velocity = $\frac{\text{Flow rate}}{\text{Area}}$

Other

$pH = -\log[H_3O^+]$

$$S^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

$$t = \frac{\bar{x_1} - \overline{x_2}}{\sqrt{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)}}$$

$UCL_{1-\alpha} = \bar{x} + t_{\alpha, n-1} s / \sqrt{n}$

b. Water-Related

$Q_s C_s + Q_d C_d = Q_r C_r$

$DTW_c = DTW_m - (PT * Rho)$

Top of Water Table Depression from Overlaying Liquid = Apparent Thickness x Density

$C_{EW} = \mathbf{\Sigma} (C_i/d_i) / \mathbf{\Sigma} (1/d_i)$

c. Air-Related

Note: AE (mass), E (mass/energy), HI (energy/time), T (time), C_d=concentration (dry) Q_d=flow rate (dry), C_w=concentration (wet), Q_w=flow rate (wet), M% = moisture (%)

AE [Air emissions] = E * HI * T * Capacity Factor

$E = K^* C_w^* Q_w$

$C_d = C_w^*/(1\text{-}M\%/100\%)$

$Q_d = Q_w^*/(1\text{-}M\%/100\%)$

$K = 2.596 \times 10^{-9} * (MW)$ (lb/dscf)/ppm

$$E = \sum_{\frac{i=1}{n}}^{n} E_1 = \frac{E_1 + E_2 \ldots + E_n}{n}$$